

CHE 201 Fall 2019 HW 8

Joe Bilderback

TOTAL POINTS

97.5 / 100

QUESTION 1

11 15 / 15

✓ - **0 pts** Correct

- **5 pts** no unit
- **5 pts** Q incorrect
- **2 pts** gallon of hot water incorrect
- **15 pts** no submission
- **15 pts** Wrong question, problem 4.55

QUESTION 2

2 2 15 / 15

✓ - **0 pts** Correct

- **7.5 pts** Partially wrong or incomplete solution. $T_2 = 759^\circ\text{R}$ and $T_4 = 779^\circ\text{R}$
- **15 pts** Wrong solution. $T_2 = 759^\circ\text{R}$ and $T_4 = 779^\circ\text{R}$

QUESTION 3

3 3 20 / 20

✓ - **0 pts** Correct

- **3 pts** T_3 unit in K
- **10 pts** Power output
- **3 pts** for calculate T_3 , h should also included h_5 and 6
- **10 pts** $T_3 = ?$
- **5 pts** T_3 unfinished
- **20 pts** no submission

QUESTION 4

4 4 15 / 15

✓ - **0 pts** Correct

- **3 pts** Q is negative. Heat transfer from the system
- **2 pts** Final answer should be in MW
- **5 pts** Calculation error. $Q = -0.21\text{ MW}$
- **5 pts** Delta PE and Delta KE are different than zero
- **5 pts** Wrong unit conversion. $Q = -0.21\text{ MW}$

- **15 pts** Missing or wrong answer

QUESTION 5

5 5 15 / 15

✓ - **0 pts** Correct

- **5 pts** Stage 1: $Q = m\Delta u$ (closed system energy balance) = 4204.2 kJ
- **3 pts** Stage 2: correct equation but wrong solution. $Q = +8725.5\text{ kJ}$. Heat transfer to the system.
- **5 pts** Stage 2: $Q = \Delta(m.u) - h_e\Delta m = +8725.5\text{ kJ}$ (energy balance one-exit control volume). Heat transfer to the system.
- **15 pts** Missing/wrong answer

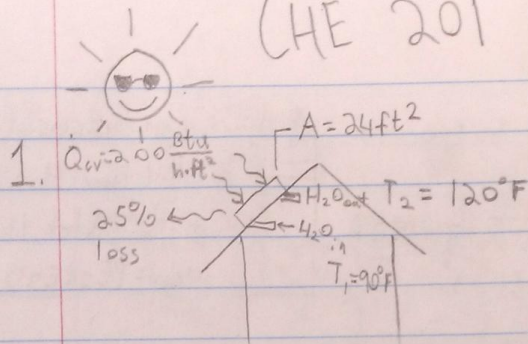
QUESTION 6

6 6 17.5 / 20

- **0 pts** Correct
- **1.5 pts** Wrong unit conversion. $m_1 = 1.2 \times 10^5\text{ kg}$ and $m_2 = 9.3 \times 10^5\text{ kg}$
- **5 pts** Wrong energy balance. Should use Eq 4.28
- **5 pts** $W = 282\text{ GJ}$
- ✓ - **2.5 pts** Work is positive (done by the compressor)
- **20 pts** Wrong answer
- **0 pts** Click here to replace this description.

Joseph Bilderback

CHE 201 HW 8



Model: Open System,
 Steady State, $\dot{W}_{cv} = PE = KE = 0$
 Incompressible fluid approx.,
 One inlet/outlet.

Find $\dot{V}_e$ Mass Balance = $\dot{m} = \dot{m}_i = \dot{m}_e$

Energy Balance

$$\frac{\Delta E}{dt} = \dot{W}_{cv} + \dot{Q}_{cv} + \dot{m} \left[h_i - h_e + \frac{V_i^2 - V_e^2}{2} + g(z_i - z_e) \right]$$

$$0 = \dot{Q}_{cv} + \dot{m}(h_i - h_e)$$

From Table A-2E

$$h_i(90^\circ) \approx 58.07 \frac{\text{Btu}}{\text{lb}} \quad h_e(120^\circ) = 88.00 \frac{\text{Btu}}{\text{lb}}$$

$$0 = 0.75 \left(\frac{200 \text{ Btu}}{\text{h} \cdot \text{ft}^2} \cdot 24 \text{ ft}^2 \right) + \dot{m} \left(58.07 \frac{\text{Btu}}{\text{lb}} - 88.00 \frac{\text{Btu}}{\text{lb}} \right)$$

$$\frac{-3600 \text{ Btu}}{\text{h}} = \dot{m} (-29.93 \frac{\text{Btu}}{\text{lb}})$$

$$\dot{m} = \frac{-3600 \text{ Btu}}{\text{h}} \cdot \frac{1 \text{ lb}}{29.93 \text{ Btu}} = \frac{120.28 \text{ lb}}{\text{h}}$$

From Table A-2E
 $\dot{V}_e \approx 0.01621 \frac{\text{ft}^3}{\text{lb}}$

$$\dot{m} = \frac{\dot{V}}{V} \quad \dot{m} = \frac{\dot{V}_e}{V_e}$$

$$\frac{120.28 \text{ lb}}{\text{h}} = \frac{\dot{V}_e}{0.01621 \frac{\text{ft}^3}{\text{lb}}} \quad \dot{V}_e = \frac{1.95 \text{ ft}^3}{\text{h}} \cdot \frac{1 \text{ gal}}{0.13368 \text{ ft}^3}$$

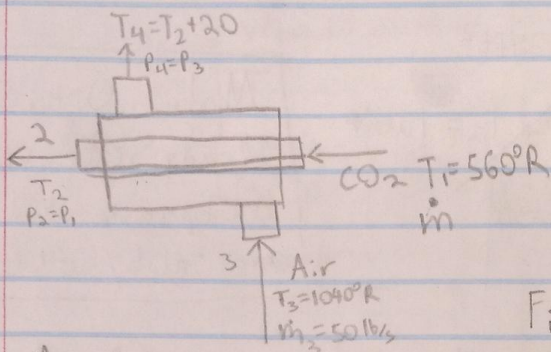
$$\dot{V}_e = 14.59 \frac{\text{gal}}{\text{h}}$$

11 15 / 15

✓ - 0 pts Correct

- 5 pts no unit
- 5 pts Q incorrect
- 2 pts gallon of hot water incorrect
- 15 pts no submission
- 15 pts Wrong question, problem 4.55

2.



Model: Open System, Steady State

Parallel Flow Heat Exchanger,

2 inlets/outlet, Ideal gases,

$$\dot{W}_{cv} = PE = KE = 0$$

Find T at exits

$$\text{Mass balance } \dot{m}_1 = \dot{m}_2 = \dot{m}_{CO_2} \quad \dot{m}_3 = \dot{m}_4 = \dot{m}_{Air}$$

$$\text{Energy Balance } 0 = \dot{m}_{CO_2}(h_1 - h_2) + \dot{m}_{Air}(h_3 - h_4)$$

$$h_2 = \dot{m}_{CO_2}(h_2 - h_1) \neq \dot{m}_{Air}$$

$$h_1 = 4235.8 \frac{\text{Btu}}{\text{lbmol}} (T_A = 23E) \quad h_1 = \frac{4235.8}{44.01} = 96.25 \frac{\text{Btu}}{\text{lb}}$$

$$h_3 = 250.95 \frac{\text{Btu}}{\text{lb}} \quad (\text{From Table A-23E})$$

$$\dot{Q}_{CO_2} = \dot{m}_{CO_2}(h_2 - h_1)$$

$$\dot{Q}_{CO_2} = (73.7 \text{ lb/s})(h_2 - 96.25 \frac{\text{Btu}}{\text{lb}})$$

$$\dot{Q}_{Air} = \dot{m}_{Air}(h_3 - h_4) \quad \dot{Q}_{Air} = 50 \text{ lb/s} (250.95 \frac{\text{Btu}}{\text{lb}} - h_4)$$

$$\text{if } T_2 = 760^\circ R \quad \& \quad T_4 = 780^\circ R$$

$$\dot{Q}_2 = 3700.79 \text{ Btu/s} \quad \& \quad \dot{Q}_4 = 3200.5 \text{ Btu/s}$$

$$T_4 = T_2 + 20 \checkmark$$

$$\boxed{T_2 = 760^\circ R \quad \& \quad T_4 = 780^\circ R}$$

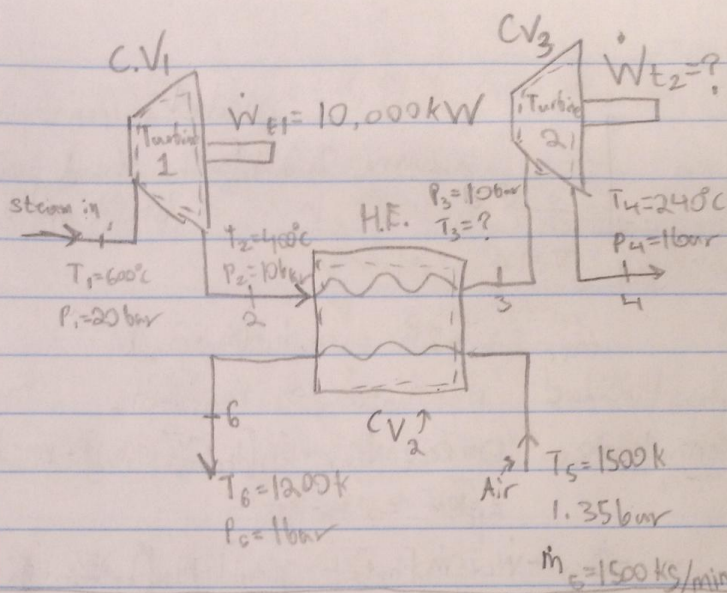
2 2 15 / 15

✓ - 0 pts Correct

- 7.5 pts Partially wrong or incomplete solution. $T_2 = 759^\circ\text{R}$ and $T_4 = 779^\circ\text{R}$

- 15 pts Wrong solution. $T_2 = 759^\circ\text{R}$ and $T_4 = 779^\circ\text{R}$

3.



- Model: open system,
Ideal gas, steady
state, $\dot{Q}_{cv} = \dot{K}E = \dot{P}E = 0$
for all cvs

a) CV_1 M balance $\dot{m}_T = \dot{m}_1 = \dot{m}_2$ From table A-4 $h_1 = 3690.1 \frac{kJ}{kg}$ $h_2 = 3263.9 \frac{kJ}{kg}$ E balance $= 0 = \dot{W}_{cv} + \dot{m}(h_1 - h_2)$

$$\dot{W}_{cv} = \dot{m}_T(h_1 - h_2) \rightarrow \dot{m}_T = \frac{\dot{W}_{cv}}{(h_1 - h_2)}$$

$$\dot{m}_T = \frac{10,000 \text{ kW}}{(3690.1 - 3263.9) \frac{kJ}{kg}} = 23.46 \text{ kg/s} = 1407.78977 \text{ kg/min}$$

 CV_2

$$\dot{m}_T = \dot{m}_3$$

From Table A-4 & A-22

$$\dot{m}_T(h_2 - h_3) + \dot{m}_5(h_5 - h_6) = 0 \quad h_2 = 3263.9 \quad h_5 = 1635.97 \frac{kJ}{kg}$$

$$1407.79(3263.9 - h_3) + 1500(1635.97 - 1277.79) = 0 \quad h_6 = 1277.79 \frac{kJ}{kg}$$

$$-1407.79 h_3 = -537270 - (1407.79 \times 3263.9)$$

$$h_3 = \frac{-5132155.781}{-1407.79} = 3645.54 \frac{kJ}{kg}$$

Interpolate from T A-4 at 10 bar

$$\frac{(600 - 540)^\circ C}{(3697.9 - 3565.6) \frac{kJ}{kg}} = \frac{(T_3 - 540)^\circ C}{(3645.54 - 3565.6) \frac{kJ}{kg}}$$

$$T_3 = \frac{60^\circ C (79.94)}{132.3 \frac{kJ}{kg}} + 540^\circ C = 576.25^\circ C = 849.4 K$$

b)

 CV_3

$$\dot{m}_3 = \dot{m}_T$$

$$0 = -\dot{W}_{t2} + \dot{m}_T(h_3 - h_4) \rightarrow \dot{W}_{t2} = \dot{m}_T(h_3 - h_4)$$

From T A-4
 $h_4 = 2954.5 \frac{kJ}{kg}$

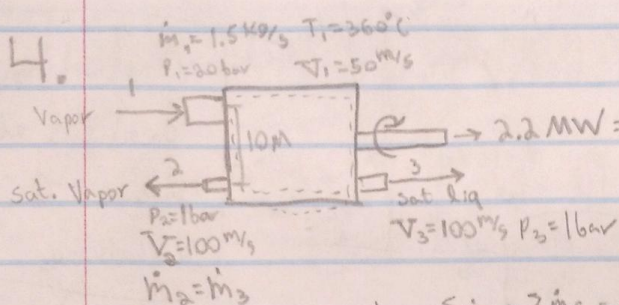
$$\dot{W}_{t2} = 23.46 \frac{kg}{s} (3645.54 - 2954.5) \frac{kJ}{kg} = 16212 \text{ kW}$$

33 20 / 20

✓ - 0 pts Correct

- 3 pts T3 unit in K
- 10 pts Power output
- 3 pts for calculate T3, h should also included h5 and 6
- 10 pts T3 = ?
- 5 pts T3 unfinished
- 20 pts no submission

4.



Model: Open System, Steady

steady state,
shaft work, 1 inlet 2 outlets

$$\dot{m}_1 = \dot{m}_2 + \dot{m}_3 \rightarrow \dot{m}_1 = 2\dot{m}_2 \quad \dot{m}_2 = 0.75 \text{ kg/s}$$

$$\text{Mass balance } \dot{m}_1 = \dot{m}_2 + \dot{m}_3 \rightarrow \dot{m}_1 = 2\dot{m}_2 \quad \dot{m}_2 = 0.75 \text{ kg/s}$$

$$\text{Energy balance } 0 = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}_1 \left[h_1 + \frac{V_1^2}{2} + g(z_1) \right] + \dot{m}_2 \left[h_2 - h_3 + \frac{V_2^2 + V_3^2}{2} + g(z_2 - z_3) \right]$$

$$z_1 = 10 \text{ m} \quad z_2 = z_3 = 0 \text{ m}$$

$$-\dot{Q}_{cv} = -\dot{W}_{cv} + \dot{m}_1 \left[h_1 + \frac{V_1^2}{2} + g(z_1) \right] + \dot{m}_2 \left[h_2 + \frac{V_2^2}{2} + g(z_2) \right] - \dot{m}_3 \left[h_3 + \frac{V_3^2}{2} + g(z_3) \right]$$

From T A-4 at 20 bar From T A-3

$$h_1 = 3159.3 \text{ kJ/kg} \quad h_2 = 2675.5 \text{ kJ/kg} \quad h_3 = 417.46 \text{ kJ/kg}$$

$$h_1 = 3159.3 \times 10^3 \text{ J/kg} \quad h_2 = 2675.5 \times 10^3 \text{ J/kg} \quad h_3 = 417.46 \times 10^3 \text{ J/kg}$$

$$-\dot{Q}_{cv} = -2.2 \times 10^6 + 1.5 \left(3159.3 \times 10^3 + \frac{50^2}{2} + 9.81 \times 10 \right) - 0.75 \left(2675.5 \times 10^3 + \frac{100^2}{2} + 0 \right) - 0.75 \left(417.46 \times 10^3 + \frac{100^2}{2} + 0 \right)$$

$$\dot{Q}_{cv} = -213752.15 \text{ W} \times \frac{1 \text{ MW}}{1000000 \text{ W}} = -0.21 \text{ MW}$$

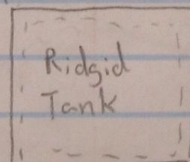
5. From T A-3 at 1 bar Stage 1:

$$X_1 = 0.01 \quad X_2 = 0.17$$

$$V_{f1} = 0.001043 \text{ m}^3/\text{kg} \quad V_{g1} = 1.6941 \text{ m}^3/\text{kg}$$

$$U_{f1} = 417.4 \text{ kJ/kg} \quad U_{g1} = 2505.6 \text{ kJ/kg}$$

$$m_1 = \frac{V_1}{V_1} = \frac{0.1 \text{ m}^3}{0.018 \text{ m}^3/\text{kg}} = 5.56 \text{ kg}$$



$$V_1 = (1-X)V_{f1} + V_{g1}X \quad V_1 = (99)0.001043 \times 10^{-3} \text{ m}^3/\text{kg} + (0.01)1.6941 \text{ m}^3/\text{kg} = 0.018 \text{ m}^3/\text{kg}$$

$$U_1 = (1-X)U_{f1} + U_{g1}X \quad U_1 = (99)417.4 \text{ kJ/kg} + (0.01)2505.6 \text{ kJ/kg} = 438.282 \text{ kJ/kg}$$

@ 20 bar

$$V_{f2} = 0.001177 \text{ m}^3/\text{kg} \quad V_{g2} = 0.099587 \text{ m}^3/\text{kg}$$

$$U_{f2} = 906.12 \text{ kJ/kg} \quad U_{g2} = 2599.1 \text{ kJ/kg}$$

$$0.018 \text{ m}^3/\text{kg} = 0.001177 \text{ m}^3/\text{kg} (1-X_2) + X_2 (0.099587 \text{ m}^3/\text{kg})$$

$$X_2 = 0.17$$

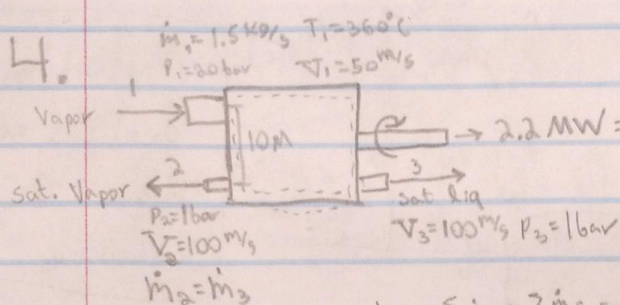
$$U_2 = U_{f2} (1-X) + U_{g2} X \rightarrow U_2 = 906.12 (1-0.17) + 2599.1 (0.17) = 1193.9266 \text{ kJ/kg}$$

4 4 15 / 15

✓ - 0 pts Correct

- 3 pts Q is negative. Heat transfer from the system
- 2 pts Final answer should be in MW
- 5 pts Calculation error. $Q = -0.21$ MW
- 5 pts Delta PE and Delta KE are different than zero
- 5 pts Wrong unit conversion. $Q = -0.21$ MW
- 15 pts Missing or wrong answer

4.



Model: Open System, Steady

state, shaft work, 1 inlet 2 outlets

$$\dot{m}_1 = \dot{m}_2 + \dot{m}_3 \rightarrow \dot{m}_1 = 2\dot{m}_2 \quad \dot{m}_2 = 0.75 \text{ kg/s}$$

$$\text{Mass balance } \dot{m}_1 = \dot{m}_2 + \dot{m}_3 \rightarrow \dot{m}_1 = 2\dot{m}_2 \quad \dot{m}_2 = 0.75 \text{ kg/s}$$

$$\text{Energy balance } 0 = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}_1 \left[h_1 + \frac{V_1^2}{2} + g(z_1) \right] + \dot{m}_2 \left[h_2 - h_3 + \frac{V_2^2 + V_3^2}{2} + g(z_2 - z_3) \right]$$

$$z_1 = 10 \text{ m} \quad z_2 = z_3 = 0 \text{ m}$$

$$-\dot{Q}_{cv} = -\dot{W}_{cv} + \dot{m}_1 \left[h_1 + \frac{V_1^2}{2} + g(z_1) \right] + \dot{m}_2 \left[h_2 + \frac{V_2^2}{2} + g(z_2) \right] - \dot{m}_3 \left[h_3 + \frac{V_3^2}{2} + g(z_3) \right]$$

From T A-4 at 20 bar From T A-3

$$h_1 = 3159.3 \text{ kJ/kg} \quad h_2 = 2675.5 \text{ kJ/kg} \quad h_3 = 417.46 \text{ kJ/kg}$$

$$h_1 = 3159.3 \times 10^3 \text{ J/kg} \quad h_2 = 2675.5 \times 10^3 \text{ J/kg} \quad h_3 = 417.46 \times 10^3 \text{ J/kg}$$

$$-\dot{Q}_{cv} = -2.2 \times 10^6 + 1.5 \left(3159.3 \times 10^3 + \frac{(50)^2}{2} + 9.81 \times 10 \right) - 0.75 \left(2675.5 \times 10^3 + \frac{(100)^2}{2} + 0 \right) - 0.75 \left(417.46 \times 10^3 + \frac{(100)^2}{2} + 0 \right)$$

$$\dot{Q}_{cv} = -213752.15 \text{ W} \times \frac{1 \text{ MW}}{1000000 \text{ W}} = -0.21 \text{ MW}$$

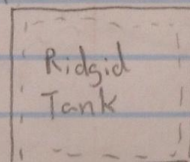
5. From T A-3 at 1 bar Stage 1:

$$x_1 = 0.01 \quad x_2 = 0.17$$

$$V_{f1} = 0.001043 \text{ m}^3/\text{kg} \quad V_{g1} = 1.6941 \text{ m}^3/\text{kg}$$

$$u_{f1} = 417.4 \text{ kJ/kg} \quad u_{g1} = 2505.6 \text{ kJ/kg}$$

$$m_1 = \frac{V_1}{V_1} = \frac{0.1 \text{ m}^3}{0.018 \text{ m}^3/\text{kg}} = 5.56 \text{ kg}$$



$$V_1 = (1-x)V_{f1} + V_{g1}x \quad V_1 = (99)1.043 \times 10^{-3} \text{ m}^3/\text{kg} + (0.01)1.6941 \text{ m}^3/\text{kg} = 0.1018 \text{ m}^3/\text{kg}$$

$$u_1 = (1-x)u_{f1} + u_{g1}x \quad u_1 = (99)417.4 \text{ kJ/kg} + (0.01)2505.6 \text{ kJ/kg} = 438.282 \text{ kJ/kg}$$

@ 20 bar

$$V_{f2} = 0.001177 \text{ m}^3/\text{kg} \quad V_{g2} = 0.099587 \text{ m}^3/\text{kg}$$

$$u_{f2} = 906.12 \text{ kJ/kg} \quad u_{g2} = 2599.1 \text{ kJ/kg}$$

$$0.1018 \text{ m}^3/\text{kg} = 0.001177 \text{ m}^3/\text{kg} (1-x_2) + x_2 (0.099587 \text{ m}^3/\text{kg})$$

$$x_2 = 0.17$$

$$u_2 = u_{f2} (1-x) + u_{g2} x \rightarrow u_2 = 906.12 (1-0.17) + 2599.1 (0.17) = 1193.9266 \text{ kJ/kg}$$

$$Q_1 = m(u_2 - u_1) = 5.56 \text{ kg} (1193.93 - 438.262) \frac{\text{kJ}}{\text{kg}} = \boxed{4201.4 \text{ kJ}}$$

From stage 2:

Table A-3 $v_3 = 0.099587 \text{ m}^3/\text{kg}$ $u_3 = 2599.1 \text{ kJ/kg}$ $h_3 = 2798.3 \text{ kJ/kg}$

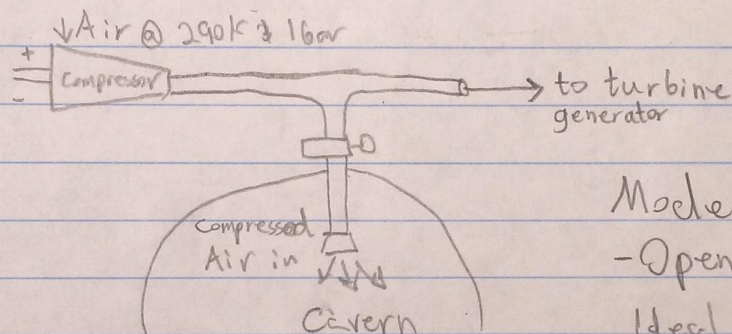
$$m_3 = \frac{V}{v} = \frac{0.1 \text{ m}^3}{0.099587 \text{ m}^3/\text{kg}} = 1.004 \text{ kg}$$

$$\dot{Q}_2 = (m_3 u_3 - m_2 u_2) - h_3 (m_3 - m_2)$$

$$(1.004 \text{ kg} \cdot 2599.1 \frac{\text{kJ}}{\text{kg}} - 5.56 \text{ kg} \cdot 1193.93 \frac{\text{kJ}}{\text{kg}}) - 2798.3 \frac{\text{kJ}}{\text{kg}} (1.004 - 5.56)$$

$$= \boxed{8720.3 \text{ kJ/kg}}$$

6.



Model:

- Open system, Air is
Ideal gas, $\dot{Q}_{cv} = \dot{A}KE = \dot{A}PE = 0$

a) $pV = mRT$

$$m_1 = \frac{p_1 V}{RT_1}$$

$$\frac{(1 \text{ bar})(10^5 \text{ m}^3) \times 100 \text{ kPa}}{8.314 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}} \cdot 2897 \frac{\text{kg}}{\text{kmol}} \cdot 290 \text{ K}} = \boxed{12019.6 \text{ kg}}$$

$$p_2 V = m_2 R T_2 \quad \therefore \quad m_2 = \frac{p_2 V}{R T_2} = \frac{(2 \text{ bar})(10^5 \text{ m}^3) \times 100 \text{ kPa}}{8.314 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}} (2897 \frac{\text{kg}}{\text{kmol}}) (790 \text{ K})}$$

$$m_2 = \boxed{926255.24 \text{ kg}}$$

b) $\frac{dU_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}_1 h_1 - \dot{m}_2 h_2$

$$\frac{dU_{cv}}{dt} = -\dot{W}_{cv} + \dot{m}_1 h_1 - 0$$

$$\Delta U_{cv} = -\dot{W}_{cv} + \int \dot{m}_1 h_1 dt \quad m_2 u_2 - m_1 u_1 = -W_{cv} + h_1 (m_2 - m_1)$$

$$W_{cv} = m_2 (h_1 - u_2) + m_1 (u_1 - h_1)$$

5 5 15 / 15

✓ - 0 pts Correct

- 5 pts Stage 1: $Q = m\Delta u$ (closed system energy balance) = 4204.2 kJ

- 3 pts Stage 2: correct equation but wrong solution. $Q = +8725.5$ kJ. Heat transfer to the system.

- 5 pts Stage 2: $Q = \Delta(m.u) - h_e\Delta m = +8725.5$ kJ (energy balance one-exit control volume). Heat transfer to the system.

- 15 pts Missing/wrong answer

$$Q_1 = m(u_2 - u_1) = 5.56 \text{ kg} (1193.93 - 438.262) \frac{\text{kJ}}{\text{kg}} = \boxed{4201.4 \text{ kJ}}$$

From stage 2:

Table A-3 $v_3 = 0.099587 \text{ m}^3/\text{kg}$ $u_3 = 2599.1 \text{ kJ/kg}$ $h_3 = 2798.3 \text{ kJ/kg}$

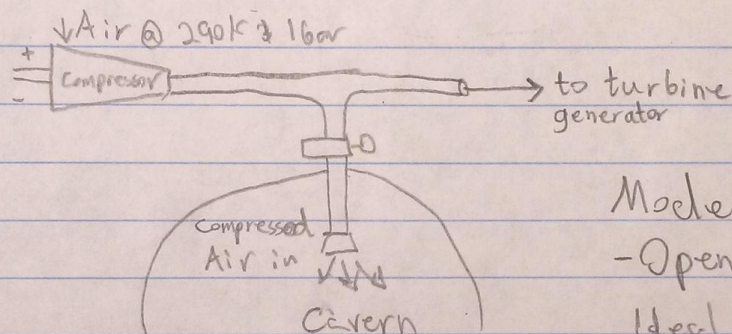
$$m_3 = \frac{V}{v} = \frac{0.1 \text{ m}^3}{0.099587 \text{ m}^3/\text{kg}} = 1.004 \text{ kg}$$

$$\dot{Q}_2 = (m_3 u_3 - m_2 u_2) - h_3 (m_3 - m_2)$$

$$(1.004 \text{ kg} \cdot 2599.1 \frac{\text{kJ}}{\text{kg}} - 5.56 \text{ kg} \cdot 1193.93 \frac{\text{kJ}}{\text{kg}}) - 2798.3 \frac{\text{kJ}}{\text{kg}} (1.004 - 5.56)$$

$$= \boxed{8720.3 \text{ kJ/kg}}$$

6.



Model:

- Open system, Air is
Ideal gas, $\dot{Q}_{cv} = \dot{A}KE = \dot{A}PE = 0$

a) $pV = mRT$

$$m_1 = \frac{p_1 V}{RT_1}$$

$$\frac{(1 \text{ bar})(10^5 \text{ m}^3) \times 100 \text{ kPa}}{8.314 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}} \cdot 2897 \frac{\text{kg}}{\text{kmol}} \cdot 290 \text{ K}} = \boxed{12019.6 \text{ kg}}$$

$$p_2 V = m_2 R T_2 \quad \therefore \quad m_2 = \frac{p_2 V}{R T_2} = \frac{(2 \text{ bar})(10^5 \text{ m}^3) \times 100 \text{ kPa}}{(8.314 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}})(2897 \frac{\text{kg}}{\text{kmol}})(790 \text{ K})}$$

$$m_2 = \boxed{926255.24 \text{ kg}}$$

b) $\frac{dU_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}_1 h_1 - \dot{m}_2 h_2$

$$\frac{dU_{cv}}{dt} = -\dot{W}_{cv} + \dot{m}_1 h_1 - 0$$

$$\Delta U_{cv} = -\dot{W}_{cv} + \int \dot{m}_1 h_1 dt \quad m_2 u_2 - m_1 u_1 = -W_{cv} + h_1 (m_2 - m_1)$$

$$W_{cv} = m_2 (h_1 - u_2) + m_1 (u_1 - h_1)$$

Table A-22 @ 290 K $u_i = 206.91 \text{ kJ/kg}$ $h_i = 290.16 \text{ kJ/kg}$

@ 790 K $u_2 = 584.21 \text{ kJ/kg}$

$$W_{cv} = m_2(h_i - u_2) + m_1(u_i - h_i)$$

$$\frac{1}{10^6} \times [(92625524)(290.16 - 584.21) + 120154.6(206.91 - 290.16)]$$

$$[-282.36 \text{ GJ}]$$

6 6 17.5 / 20

- 0 pts Correct
- 1.5 pts Wrong unit conversion. $m_1 = 1.2 \times 10^5 \text{ kg}$ and $m_2 = 9.3 \times 10^5 \text{ kg}$
- 5 pts Wrong energy balance. Should use Eq 4.28
- 5 pts $W = 282 \text{ GJ}$
- ✓ - 2.5 pts Work is positive (done by the compressor)
- 20 pts Wrong answer
- 0 pts [Click here to replace this description.](#)